

Preliminary Conceptual Design of a Ground-Based LEO Tracking Radar

1. Objective & Requirements

The objective is to develop a preliminary ground-based monostatic radar architecture capable of detecting and continuously tracking Low Earth Orbit (LEO) objects between 500 km and 800 km altitude, including a metallic spherical target of approximately 10 cm diameter.

Requirement	Target
Operating altitude	500-800 km
Target size	~10 cm
Detection margin	≥10 dB
Range resolution	≤10 m
Velocity accuracy	<1 m/s

2. Design Approach

The problem was decomposed into: mission requirements -> frequency selection -> waveform -> link budget -> antenna -> transmitter -> receiver -> signal processing -> tracking -> orbit determination. The 800 km case is used as the primary design point because it is the most demanding range.

3. Proposed System Architecture

Waveform Generator -> RF Exciter/Driver -> 15 kW PA -> T/R Network -> 16 m Parabolic Antenna -> LEO Target -> Receiver/LNA/Downconversion -> ADC/FPGA/DSP -> Range/Doppler/Angle Processing -> Detection & Tracking -> Orbit Determination

4. Baseline Design

Parameter	Preliminary design
Frequency	3 GHz
Wavelength	0.1 m
Waveform	LFM chirp
Bandwidth	15 MHz
Pulse width	100 μs
Peak RF power	15 kW
Antenna	16 m parabolic reflector
Antenna efficiency	65% assumed
Nominal target RCS	0.00785 m²
Design range	800 km

Range resolution: $\Delta R = c/(2B) = 3 \times 10^8 / [2(15 \times 10^6)] = 10 \text{ m}$. The proposed waveform therefore meets the specified range-resolution requirement.

Design philosophy: The architecture prioritizes a traceable system-level solution over premature component-level optimization. Assumptions are explicitly identified and key parameters are tested through link-budget and sensitivity analysis.

RF, Antenna, Transmitter & Receiver

5. Frequency and Waveform Selection

3 GHz is selected as a preliminary operating frequency. It corresponds to $\lambda = 0.1$ m and provides a practical compromise between antenna aperture, angular resolution, Doppler sensitivity, RF hardware practicality and atmospheric propagation.

An LFM chirp is selected to obtain the required range resolution while allowing a relatively long pulse for improved energy-on-target. $B = 15$ MHz, $\tau = 100$ μ s, giving $BT = 1500$ and $\Delta R = 10$ m.

6. Antenna Design

A 16 m parabolic reflector is proposed for both transmission and reception. With $\eta = 65\%$, $G = \eta(\pi D/\lambda)^2 \approx 52.2$ dBi. The approximate half-power beamwidth is $\theta_{HPBW} \approx 70\lambda/D \approx 0.44^\circ$. A two-axis azimuth/elevation mount with absolute encoders and closed-loop servo control is proposed.

A monopulse angle-measurement architecture is preferred for instantaneous angular-error estimation. A preliminary angular measurement target of $\leq 0.05^\circ$ is adopted for subsequent detailed antenna, feed and calibration analysis.

7. Transmitter Design

The preliminary transmitter provides 15 kW peak RF output using a modular PA architecture. Multiple solid-state/GaN PA modules can be combined to achieve the required power. A preliminary PA efficiency of approximately 60% is assumed for power and thermal estimates.

Further engineering is required for power combining, thermal management, spurious emissions, impedance matching, protection, monitoring, timing and redundancy.

8. Receiver Design

Proposed chain: Antenna -> T/R protection -> RF filtering -> LNA -> mixer/downconversion -> IF filtering/amplification -> ADC -> FPGA/DSP. A preliminary receiver noise figure of 3 dB is assumed. The receiver supports coherent reception, pulse compression, Doppler processing, CFAR, monopulse angle estimation and track processing.

9. Key RF Parameters

<i>Parameter</i>	<i>Baseline</i>
<i>Frequency</i>	3 GHz
<i>Wavelength</i>	0.1 m
<i>Bandwidth</i>	15 MHz
<i>Pulse width</i>	100 μ s
<i>Time-bandwidth product</i>	1500
<i>Peak RF power</i>	15 kW
<i>Antenna diameter</i>	16 m
<i>Antenna gain</i>	~52.2 dBi
<i>HPBW</i>	~0.44°
<i>Receiver NF</i>	3 dB assumed
<i>Range resolution</i>	10 m

Link Budget, Detection & Doppler Processing

10. Detection Link Budget

The 800 km case is evaluated as the worst-case design point. For a 10 cm metallic sphere, the nominal RCS is approximated as $\sigma = \pi(0.05)^2 = 0.00785 \text{ m}^2$.

Using the monostatic radar equation with the baseline parameters gives an estimated received echo of approximately -135.8 dBm.

Quantity	Value
Design range	800 km
Nominal RCS	0.00785 m ²
Received echo	~-135.8 dBm
Receiver noise	~-99.2 dBm
Raw SNR	~-36.6 dB
Pulse compression gain	~31.8 dB
100-pulse coherent gain	20 dB
Ideal processed SNR	~15.2 dB
Implementation allowance	5 dB
Effective preliminary margin	~10.2 dB

The receiver noise estimate uses KTB with a 15 MHz bandwidth and a 3 dB assumed noise figure. After pulse compression and 100-pulse coherent integration, the ideal processed SNR is ~15.2 dB. Allocating 5 dB for implementation/processing losses leaves ~10.2 dB effective preliminary margin.

11. Doppler and Radial Velocity

Doppler frequency is related to radial velocity by $f_D = 2v_r/\lambda$. At $\lambda = 0.1 \text{ m}$, $f_D = 20v_r$ (Hz for v_r in m/s).

With approximately 100 coherent pulses and a reference PRF near the 800 km unambiguous-range limit ($\approx 187.5 \text{ Hz}$), the CPI is $\approx 0.533 \text{ s}$. Theoretical velocity resolution is $\Delta v \approx \lambda/(2\text{TCPI}) \approx 0.094 \text{ m/s}$, below the 1 m/s requirement.

12. PRF and Range-Doppler Ambiguity

For 800 km unambiguous range, the simple maximum PRF is approximately 187.5 Hz. Such a low PRF creates Doppler ambiguity for LEO velocities. The operational design therefore proposes staggered/multiple-PRF processing for ambiguity resolution. The exact PRF set would be optimized during detailed waveform scheduling.

13. Detection Processing Chain

Received Echo -> Digital Downconversion -> Matched Filter/Pulse Compression -> Range Bins -> Coherent Integration -> Doppler FFT -> Range-Doppler Map -> CFAR -> Detection -> Track Initiation

The weak raw echo is therefore detected through the combined effects of large aperture, transmit power, pulse compression, coherent integration and Doppler processing.

Tracking, Orbit Determination, Sensitivity & Conclusion

14. Angular Tracking

The 16 m reflector provides an approximate 0.44° HPBW. A two-axis azimuth/elevation mount with closed-loop servo control is proposed. Monopulse angle measurement provides instantaneous angular error relative to boresight. A preliminary $\leq 0.05^\circ$ measurement target is adopted for further antenna, feed and calibration analysis.

Tracking loop: Detection -> Track initiation -> Range/Doppler/Angle measurement -> Tracking filter -> Predicted target position -> Az/El servo controller -> Antenna.

15. Orbit Determination Support

Each observation can provide a time-tagged measurement vector $z_k = [R, \dot{R}, Az, El]$. Repeated observations are transformed to an Earth-referenced coordinate system and used to estimate the six-state vector $[x, y, z, v_x, v_y, v_z]^T$. An Extended Kalman Filter (EKF) is proposed for state prediction and measurement update.

A single observation is not assumed to determine a complete orbit; rather, a time history of measurements over a suitable observation arc supports orbit estimation and prediction.

16. Sensitivity Analysis

<i>Parameter variation</i>	<i>Approx. margin</i>
<i>Baseline</i>	10.2 dB
<i>Range = 700 km</i>	~12.5 dB
<i>Range = 500 km</i>	~18.4 dB
<i>RCS = 0.5× nominal</i>	~7.2 dB
<i>RCS = 2× nominal</i>	~13.2 dB
<i>Tx power = 10 kW</i>	~8.4 dB
<i>Tx power = 20 kW</i>	~11.5 dB
<i>Antenna = 14 m</i>	~8.1 dB
<i>Antenna = 18 m</i>	~12.2 dB

The analysis shows $Pr \propto Pt Gt Gr \sigma / R^4$ and, for the same transmit/receive aperture, approximately $Pr \propto D^4$. Range and aperture therefore have particularly strong effects, while target RCS is a significant uncertainty.

17. Final Preliminary Specifications

<i>Parameter</i>	<i>Value</i>
<i>Frequency</i>	3 GHz
<i>Waveform</i>	15 MHz LFM, 100 μ s
<i>Peak power</i>	15 kW
<i>Antenna</i>	16 m, ~52.2 dBi
<i>Range resolution</i>	10 m
<i>Velocity resolution</i>	~0.094 m/s
<i>Tracking</i>	Monopulse Az/El
<i>Detection processing</i>	Pulse compression + coherent integration + CFAR
<i>Orbit estimation</i>	Time-tagged measurements + EKF
<i>Effective preliminary margin</i>	~10.2 dB

18. Limitations & Conclusion

This is a preliminary conceptual design. Detailed antenna/feed characterization, EM simulation, high-power PA and thermal design, T/R protection, PRF scheduling, tracking dynamics, atmospheric propagation, clutter/interference analysis and operational orbit-determination validation remain to be completed.

The proposed 3 GHz, 15 kW, 16 m radar architecture provides a traceable preliminary solution for the stated LEO detection and tracking problem. Under the stated assumptions, the design achieves 10 m range resolution, theoretical velocity resolution better than 1 m/s, and approximately 10.2 dB effective detection margin at 800 km. The sensitivity study identifies antenna aperture, range and target RCS as important design drivers.